Portfolio

Julio Barriga
www.juliobarriga.com
juliodbarriga@gmail.com
+1 (615) 635-1956
+52 (452) 107-6485

3D Printed Ankle-Foot Orthoses

August 2017 - August 2020

Objective
Translate traditional fabrication method of O&P to a fully digital one using 3D technologies as 3D printing and 3D scanning.

Technical Details
We started 3D scanning with an Xbox Kinect, and now with an iPhone, moved from printing with FDM and a thickness of 8mm to SLS with a thickness of 2mm.
Modeled on Fusion 360 and Meshmixer.

Results/Outcomes
Deliver +30 functional devices in 3 different 3D printing technologies (FDM, SLA, SLS, MJF) and reduced time by 70% and costs by 50%.
Collaborated with multiple universities, hospitals, and companies, including **Formlabs.**

Clear Mask

May 2020

Objective
Create an inclusive 3D printed mask, that allows deaf people read lips through a transparent window. Also this allows to everyone to see face expressions while talking while wearing a mask.

Technical Details
The mask was designed on top of a Face Scan on Fusion 360 and 3D printed with Clear resin on a Form 3, as for the window, a 0.5mm PETG sheet was used.

Results/Outcomes
We got a good fit, and little to no fog on the window, also the mask had vents on the cheek sides to allow CO2 out of the mask.

Face Shield

April 2020

Objective
Create a PETG face shield for health workers, protecting from all directions against COVID19 sprays.

Technical Details
We used PETG 0.5mm sheet, using die casting for fast fabrication.

Results/Outcomes
Over 5k face shields donated to health workers.

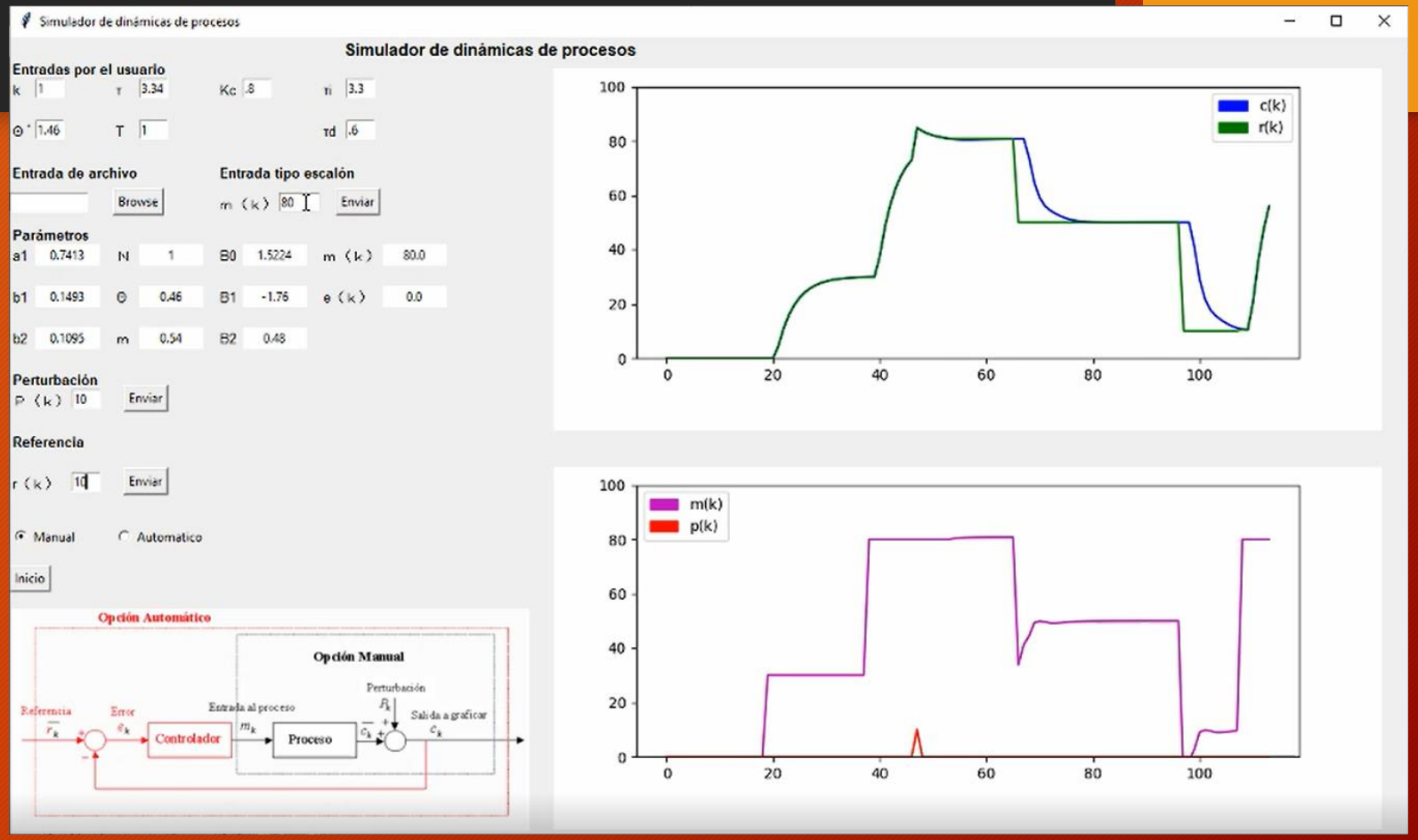
Dynamic Plant Simulator

June 2020

Objective:
Simulate a dynamic plant behavior in Python, adding an option to switch between Manual (open-loop) and Automatic (closed-loop) mode.

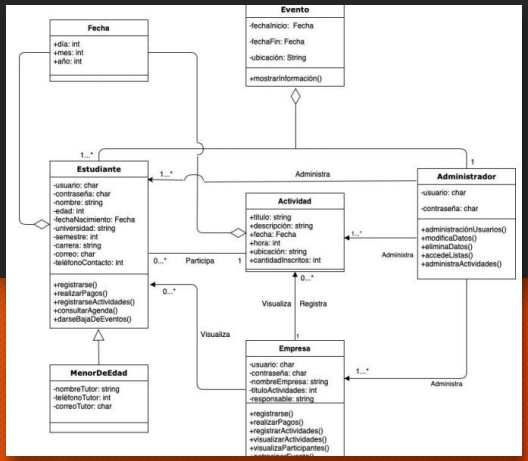
Technical Details:
To develop the simulator I used Python 3, Anaconda Navigator, Visual Studio 2019, Tkinter and matplotlib

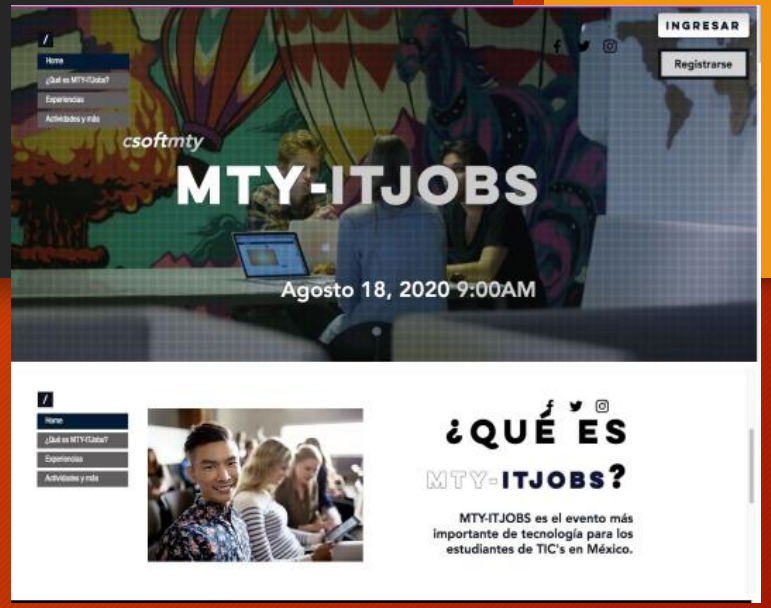
Results/Outcomes:
The result was a fully functional GUI, with 2 dynamic graphs, data entry boxes to enter the desired working parameters and a text box to display the current plant value.

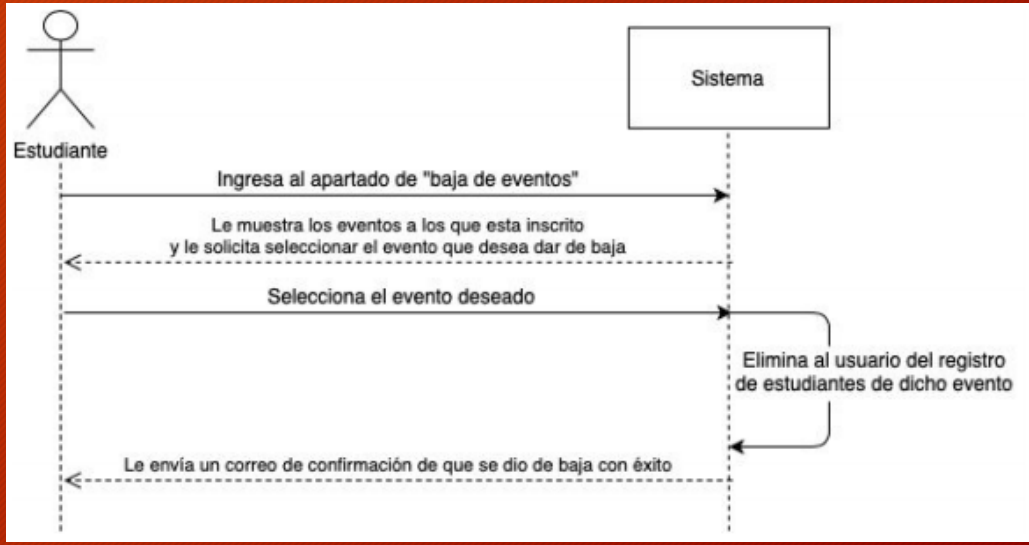


Software Engineering

June 2020







Objective

The objective was to design a website for a IT students summit “MTY-ITJOBS” based on the organizer requirements.

Technical Details

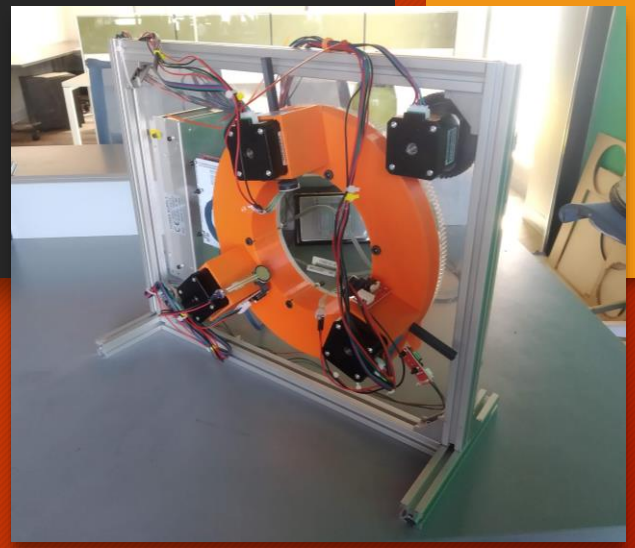
As this wasn’t a programming project but a design one, the tools used for this project was the use of SRS, Use cases, UML and test cases.
As a prototype we used GoDaddy.

Results/Outcomes

A simple and functional prototype made in GoDaddy, all the diagrams and use cases can be improved when programmed.

Residual Limb Sensing Ring

Nov 2019



Objective

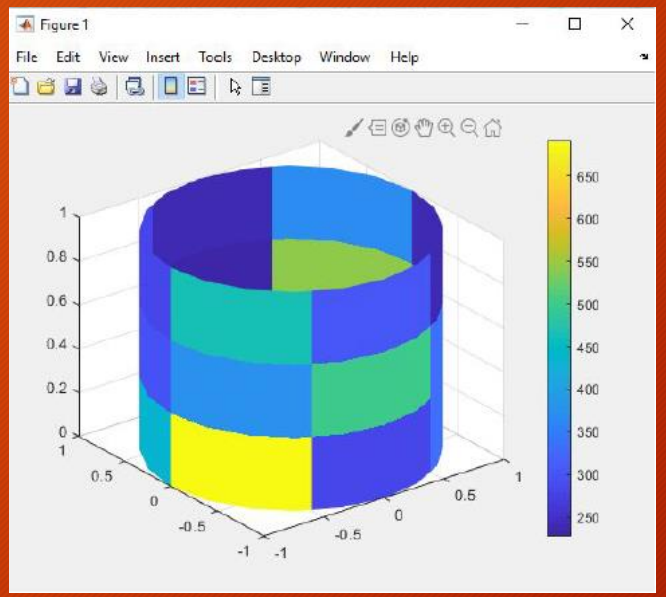
Develope a low-cost limb sensing ring prototype for residual limb tissue in amputees. Based on the FIT Socket from MIT Biomechatronic group.
The objective of the ring was to collect data of the hardness of the tissue and improve 3D printing of prosthetic limb sockets.

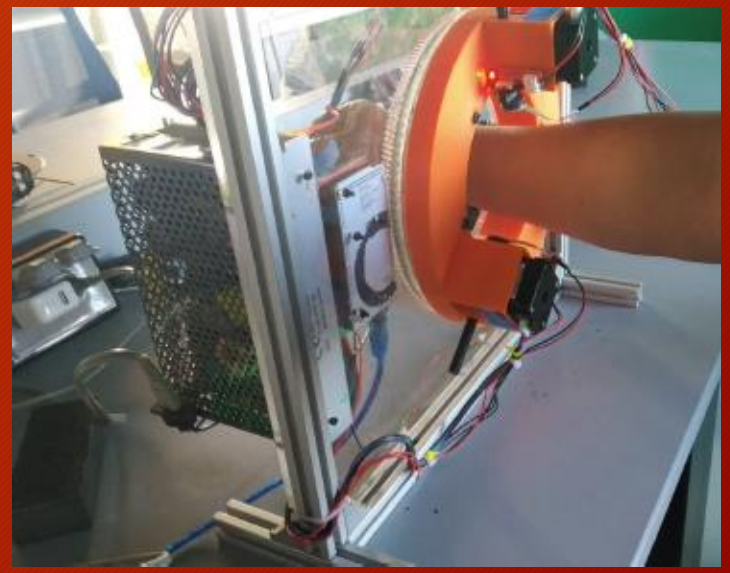
Technical Details

For this prototype, we decided to focus on upper limbs and created a small ring.
The model was designed in Fusion 360 and fabricated with 3D printing and laser cut acrylic.
For the electronics we used 4 NEMA17 stepper motors with DRV8825 drivers and an Arduino Mega; for the tissue sensors we used 3 Force Sensors FSR402.
MATLAB for data visualization.

Results/Outcomes

A functional prototype under $200 USD.
Video of the prototype





Umbrella Pavilion

Sep 2019

Objective
Design and build a deployable structure to be part of "Cielo Magico Festival". The structure had to be waterproof, windproof and be able to stand for days.

Technical Details
We used ½" and ¾" aluminum circular tubes for the links, and steel bars with ¼" bolts for the joints.
Also lycra fabric on top, to protect from direct sunlight and to give a better look to the structure.
The weight of the structure was less than 60kg, and we built it from 0 to 1 in less than 3 weeks.

Results/Outcomes
The result was good, the structure fell due to a severe storm in the first night, but only one link had to be replaced, we added 4 extra steel cables from the ground to the upper corners to avoid future accidents during the festival.

PLA with carbon nanotubes filament

May 2019

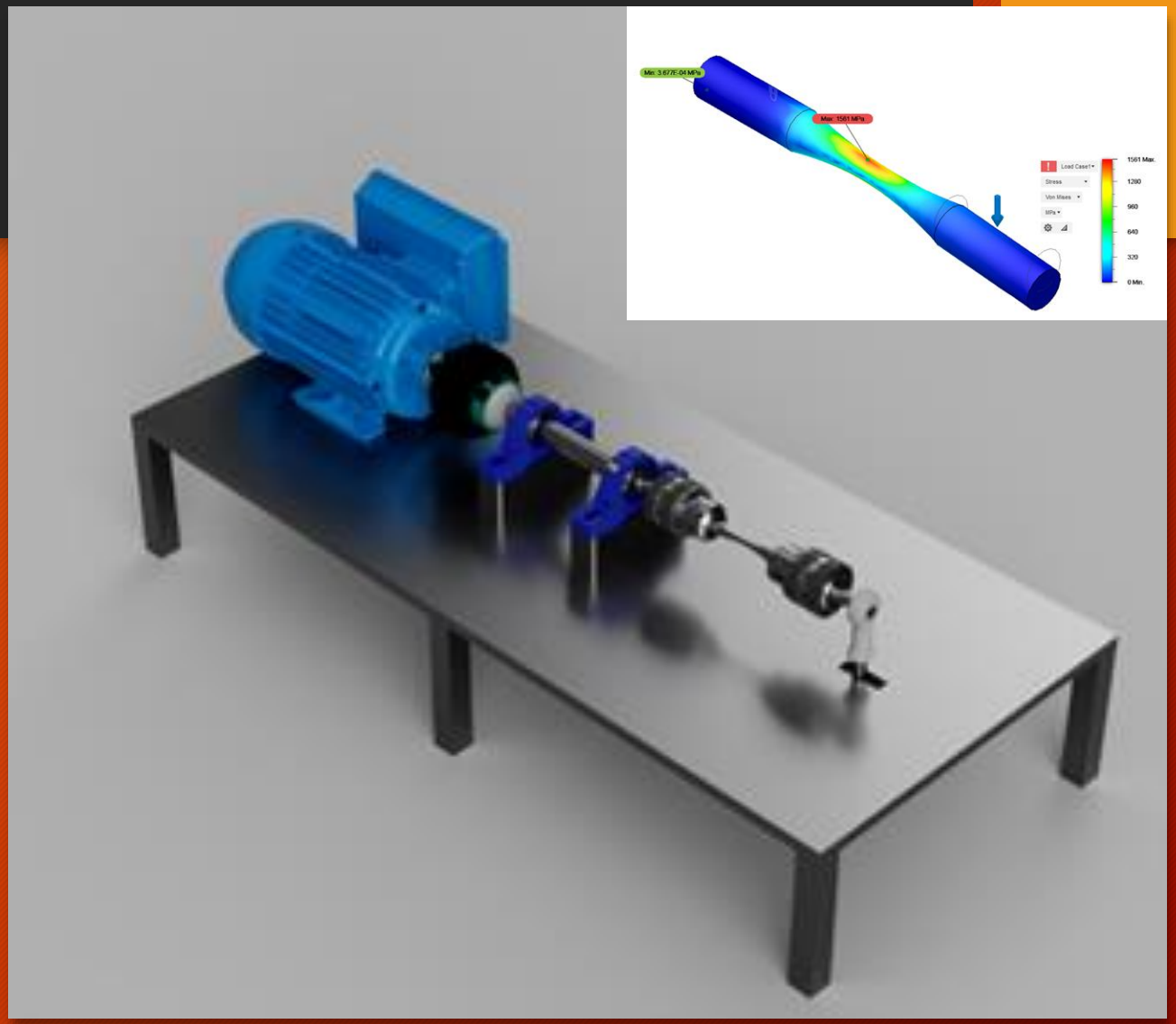
Objective
Develop a PLA with carbon nanotubes composite filament, to improve mechanical properties for 3D printed prosthetics and orthotics.

Technical Details
Conducted tensile strength tests on 3D printed specimens made with the composite following ASTM D638 and ASTM F2971 standards.

Results/Outcomes
We got good results improving by +10% the mechanical properties over regular PLA filament.

Fatigue Testing Machine

May 2019



Objective
Design a cantilever-type rotary bending testing machine.

Technical Details
We calculated the materials according with ASTM fatigue testing standards, and designed a testing machine with:
S_{ut} = 168ksi = 1160MPa
S_y = 152ksi = 1050MPa
Designed the CAD on Fusion 360 and computed FEA on relevant parts.

Results/Outcomes
Created a full material's list and created a full assembly on Fusion 360.

Transport & Process Line PLC Control

Nov 2018

Objective

Program and simulate a Transport & Process Line PLC Control.

Technical Details

For this project we used:

Software:

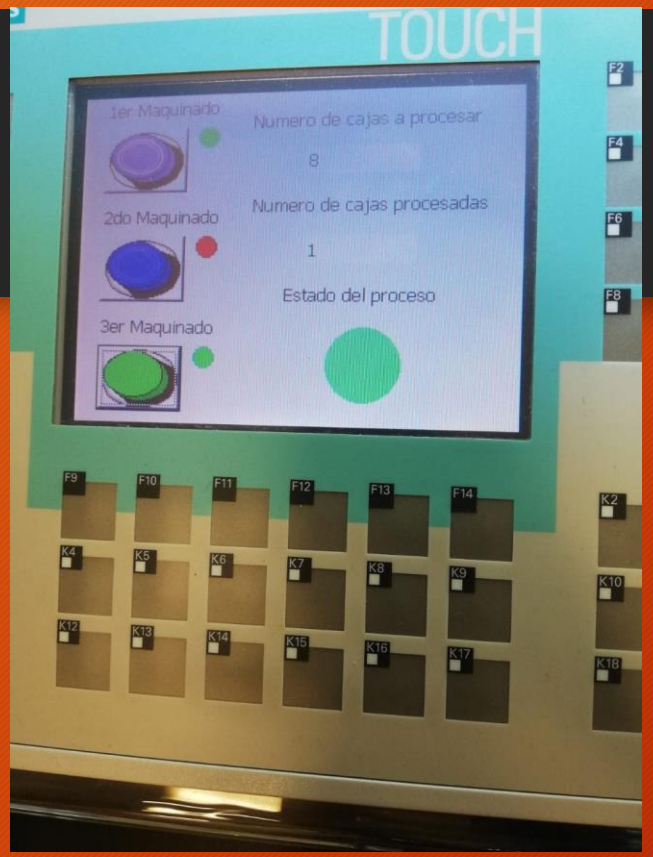
- SIMATIC Manager
- WinCC
- CRM Process

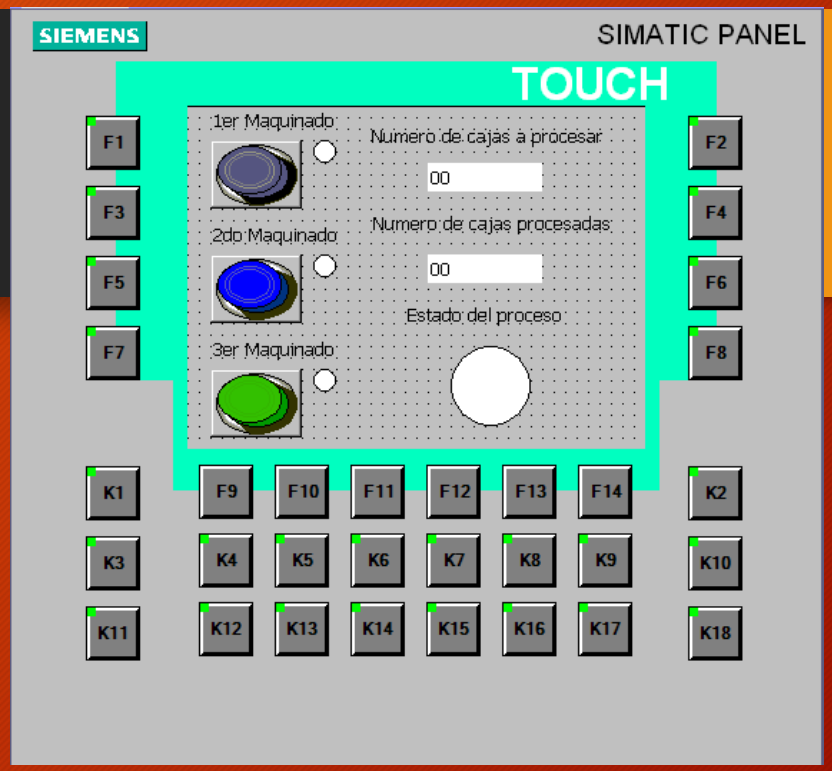
Hardware:

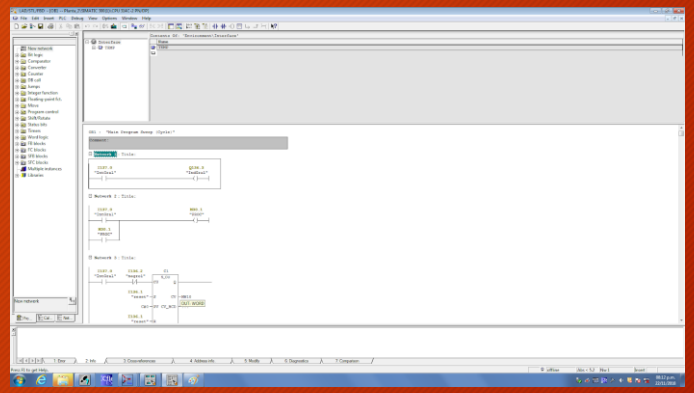
- CPU 314-2 PN/DP
- HMI OP177B PN/DP

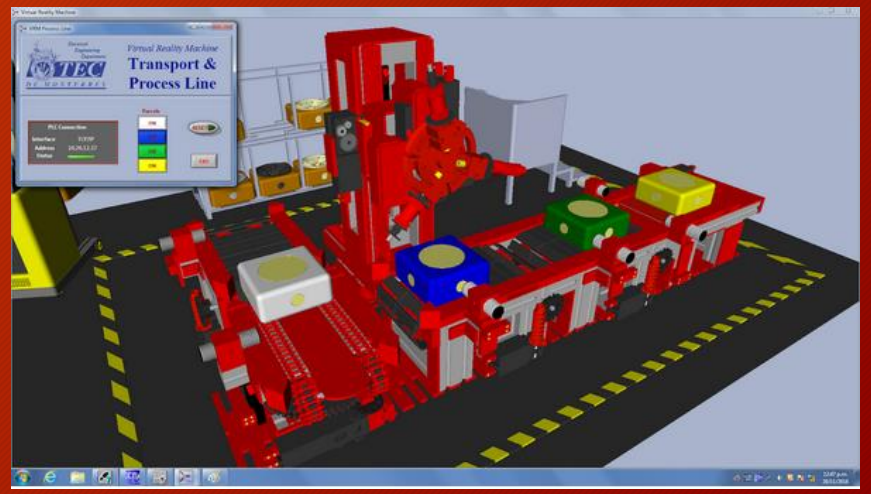
Results/Outcomes

A functional simulation, with a physical HMI working smoothly.









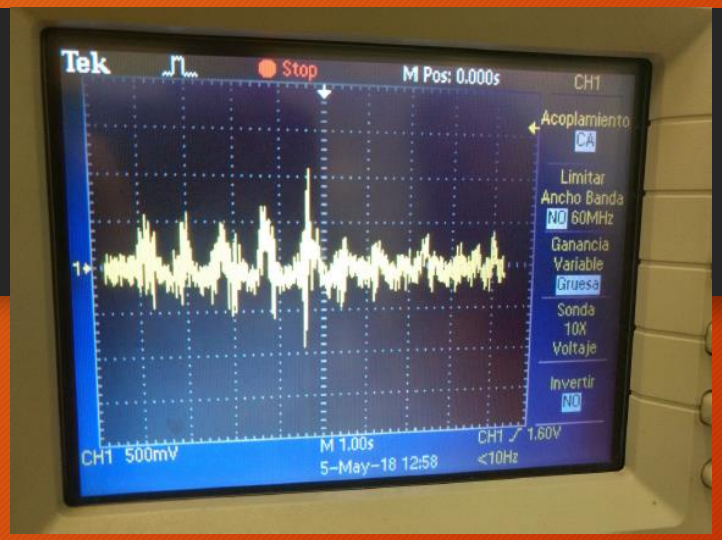
Muscle-music

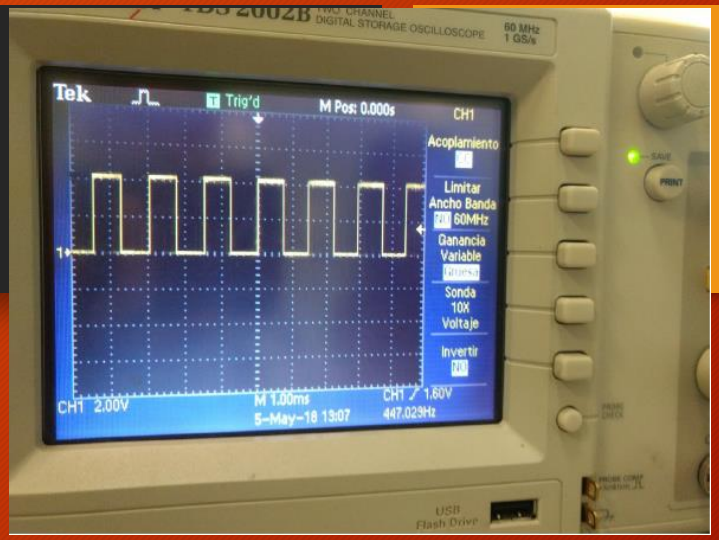
May 2018

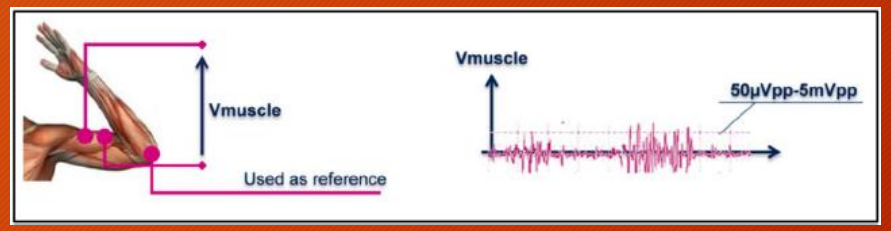
Objective
Design and develop a signal conditioning circuit to filter the muscle signal obtained from electrodes and process it with a PIC16F877A to generate a frequency to play sounds on a speaker.

Technical Details
The conditioning circuit was composed by a Pre-amplification, a low-pass & high-pass filter, an amplification and final rectification, then the signal had to pe processed with the Microcontroller programmed in Assembly.

Results/Outcomes
At the end we succeed obtaining a simple C musical scale, that we could control changing the force applied to out bicep.
We posted the tutorial on Instructables, that then got featured on Arduino'd blog and Hackaday.

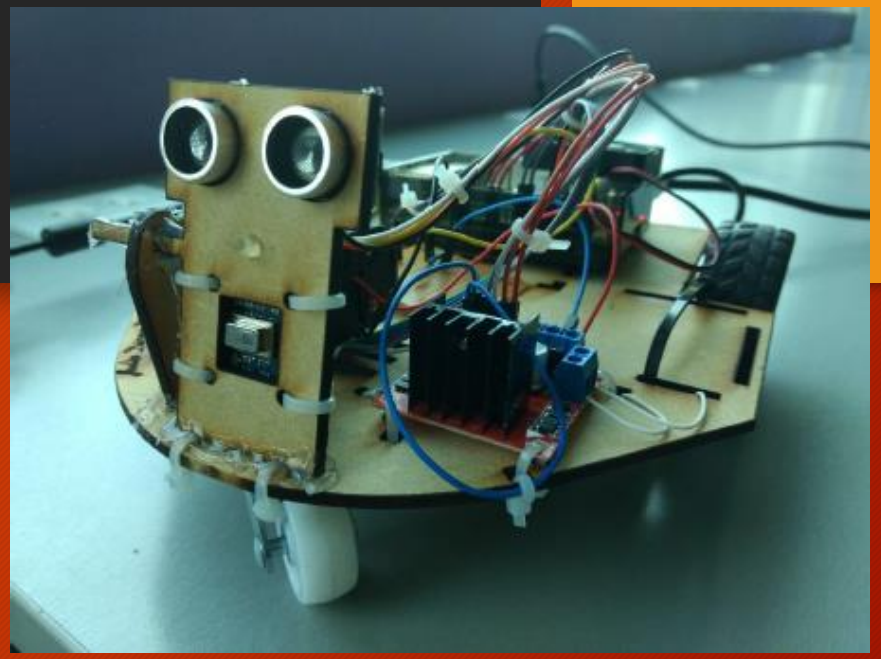






BOBOP

Nov 2017



Objective
Propose a prototype for BOBOP, The first smart robot for parents. The robot must follow a heat mass keeping ~2 meters from the subject and keep a record of the temperature of the subject.

Technical Details
To build the prototype we used 2 DC motors with an H-Bridge, and as a thermal camera Adafruit's AMG8833 sensor was used.
We used a Raspberry Pi 3 to control and keep a 2m distance from the subject.

Results/Outcomes
We got an accuracy of +-.2m in the distance kept between the robot and the subject.
The client was satisfied and chose our prototype to continue with the robot development.